Bakery Ordering UI (Cakes & Pastries)

A Project Report

23AID215 : User Interface Design

Submitted By ,

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Introduction of the Project

The project titled “Bakery Ordering UI (Cakes & Pastries)” focuses on designing and implementing a modern, visually appealing, and user-friendly web-based interface that simulates the experience of ordering bakery products online. The project aims to demonstrate User Interface Design principles by developing an interactive website which enables users to browse bakery products and customize cakes while completing orders through an easy-to-use ordering system.

Traditional bakery ordering systems present customers with multiple problems such as limited product visibility, complicated ordering processes, and lack of interactive features. The project will develop a user-friendly system through its interface design which provides users with better operational capabilities while boosting their entire experience. The system enables users to browse bakery products through structured organization and create personalized cakes while they can check their complete order before finishing the simulated payment procedure. The project creates a virtual bakery ordering system through its combination of organized layout design and interactive system elements.

The website is divided into five primary sections, which display the critical phases of the user experience. The homepage functions as the website's main entrance, which users can use to access different website sections through its interactive cake-based navigation system. The pre-made cakes and pastries page allows users to explore bakery products that are available for immediate purchase. This section also includes images of the products, along with detailed product descriptions and pricing information. The custom cake design page allows users to create their cakes by choosing their desired flavour, size, toppings and custom text while observing the cake preview update dynamically. The cart page allows the user to review their selected products and increase or decrease their quantity. The user can also see the total amount of the products selected in the cart section. The checkout page enables users to examine their cart contents, input their personal information for delivery, select a payment option and proceed with their order.

The project is developed using HTML, CSS and JavaScript, which serve as common tools for front-end web development. HTML establishes the

content structure and layout design for every webpage, CSS creates the visual design elements while maintaining interface theme consistency, and JavaScript implements interactive functions which include dynamic product filtering and cake customization preview, cart management and order confirmation simulation. The combination of these technologies enabled the building of interactive interfaces which support responsive design and adhere to modern web design standards.

The project demonstrates how effective User Interface Design principles can be used to develop an engaging and functional browsing and ordering system for a bakery website. The proposed interface provides users with a seamless ordering and browsing experience through the focus on simple design elements, clear visual presentation and interactive components exhibiting innovative user interface design specific to the bakery theme.

Individual Contributions

The project was developed as a team effort, where each member was responsible for designing and implementing a specific part of the bakery ordering and UI website. Each webpage focuses on a particular stage of the user interaction process, ensuring that the overall application is structured, interactive, and easy to navigate.

1. Pravin Raj R P : Homepage and Navigation Interface

Pravin Raj R P is responsible for designing the Homepage and Navigation Interface, which serves as the main entry point of the website. The homepage is bakery themed and provides navigation to the different sections of the website.

A distinctive feature of this page is the interactive cake-based navigation design, where different layers of the cake act as clickable links that takes the user to different parts of the website, namely the product page, custom cake designer, and checkout sections. This creative design improves user engagement while maintaining a visually appealing interface. This page also has a ‘Why Choose Us’ section, providing clear reasons to use our brand over others.

2. Aasim Mohammed M S : Pre-Made Cakes & Cart Interface

Aasim Mohammed M S is responsible for the development and implementation of the Pre-Made Cakes and Pastries Ordering Interface and Cart functionality. Using product cards with pictures, product names, descriptions, and prices, the pre-made products webpage lets users browse ready-made bakery goods. Interactive elements such as category filtering, hover effects, and order-now buttons were also included to make product selection easier and visually appealing.

Users can also review selected items, adjust their quantities, and see the total order amount in the cart section of the website. This webpage's objective was to guarantee effective cart management for the ordering process while offering a seamless product browsing experience.

3. Narendra R : Custom Cake Design Interface

Narendra R is responsible for the Custom Cake Design Interface. This section lets users design cakes according to their preferences. Users can choose various cake features for their own custom cake and place them for order from this page. Users can choose the cake type, decide the flavour, size, toppings and text to be written on the cake. The dynamic cake preview changes as users make their choices. This interactive method updates the cake design dynamically based on the features selected. Through this way users can see the final cake design in real time, making the customization process more engaging and easier to understand before ordering them.

4. Aatish Ayyapath : Checkout and Payment Interface

Aatish Ayyapath designed and implemented the Checkout and Payment Interface, which is the final step of the ordering process. On this page users can review and finalize their selected products, and then proceed to give their details and proceed with the payment. This webpage shows the order summary, section to input customer details and payment options in a clean and organized layout. This makes it simple for users to check their order before they place it. The design focuses on clarity and ease of navigation to create a smooth checkout experience.

Problem & User Description

Problem Description

In many traditional methods of ordering in bakeries, customers may find it difficult to order their required cakes by browsing, customizing their cakes, and ordering in an efficient manner. In traditional bakery ordering systems, physical bakeries may not have enough space to display their products, and customers may only be able to communicate their cake requirements by talking to the store staff. This may sometimes cause confusion regarding the design, flavor, and other customizations required in the cake.

Also in traditional ordering systems, customers are not provided with any visual interface regarding their customized cake before ordering their product.

The problem in this system is that many small-scale bakery businesses do not have any digital interface for their customers, and their websites may only be designed to display their products in a traditional manner, making it difficult for customers to use their ordering system.

The Bakery Ordering UI (Cakes & Pastries) is designed to solve the problem by creating an interactive and visually appealing interface for customers to order their required cakes in an efficient manner by browsing, customizing their cakes, and ordering in an easier way using the cart and checkout systems.

User Description

The target users of this interface are customers who wish to purchase bakery items, especially cakes and pastries online. The users could be customers who wish to purchase cakes and other bakery items for their occasions like birthdays or other events.

The interface is user-friendly, especially for customers with limited knowledge of the internet who wish to purchase items from an online website, since the features that this interface provides, such as the

navigation menu and the images that the customers can see, are very easy to understand and use by any customer, and this interface is also visually intuitive, meaning that the customers can easily understand how to use the interface to browse the items and how to customize the cakes and other items and add them to the cart.

In overall, this system aims to be simple, interesting, and friendly for customers who wish to explore and purchase bakery products online.

UI Design Explanation (Overall Project)

The user interface of the Bakery Ordering UI (Cakes & Pastries) project is designed with a focus on simplicity, clarity, and user interactivity. The design of the user interface is based on modern web interface design principles, where the user interface is designed in a manner that allows users to navigate through different sections of the website in a structured manner. It is designed with consistent design elements, navigation, and visually appealing design elements to avoid confusion for users.

One of the important features of the interface design is the structure of the layout of the website. The website is divided into different pages, each with a standardized structure for the layout. A navigation bar at the top of each page, followed by the main content of the page in the middle, and then the footer at the bottom of each page is also included. The navigation bar at the top of each page provides access to the different key pages of the website, such as the homepage, pre-made cakes page, custom cake designer, cart page and checkout.

The homepage plays a major role in the overall interface design as it is the visual entry point to the website. Rather than relying on a regular navigation menu, the homepage includes an interactive cake navigation system in which different layers of a cake are used to navigate to different webpages of the website and a ‘Why Choose Us’ to build trust with the visitors. Such an innovative approach to interface design increases user interaction while also providing a thematic connection to the idea of a bakery. The hover effects and visual effects included in the cake used for navigation also improves the interactive experience of this webpage.

Another significant element in the UI for the project is the product card layout on the pre-made cakes and pastries page. Every product on the page is placed within a structured card format consisting of the product image, product name, product description, and price. All the cards are arranged in a grid layout so that the products are properly organized according to the screen size. Various interactive features such as hover and the addition and removal buttons are also present.

Custom cake design interface is another main part of the overall UI design. This interface is where the user gets the chance to customize the cake by choosing the type of cake they want, including features like the size of the cake, preferred flavour, toppings and the text they want on the cake. A live preview is provided where the user gets a visual representation of the cake as they make the selections. The layered preview helps the user understand the changes they are making and the effect they will have on the cake. Various interface tools are provided in this part of the design as well, such as tooltips and brief descriptions.

The cart and checkout interfaces are designed in a way that clarity is achieved during the final stages of the ordering process. In the cart section, users can review their list of selected items, vary their quantity, and finally see the calculated price, which enables them to make any necessary changes to their order before proceeding to checkout. In the checkout page, customer information and preferred payment method are collected and delivery message is displayed. This step-by-step process reduces confusion and ensures that users can complete their orders smoothly.

The visual appeal of the website is based on a soft colour scheme related to the theme of a bakery. The colour scheme includes shades of chocolate brown and light cream colours along with pink highlights for interactive elements. These colours create an appealing look while being consistent with a bakery theme. Another factor taken into consideration while designing the website is the typography of the website. Decorative fonts for headings and clean fonts for body text to ensure readability have been used.

In overall, the UI design of the project combines structured layout, consistent styling, interactive elements, and responsive design practices to create an engaging and easy-to-use bakery ordering interface. By integrating visually appealing elements such as the cake-based navigation system and real-time cake customization preview, the interface provides users with an intuitive and enjoyable ordering experience.

Individual Pages (HTML / CSS / JavaScript) Explanation

The website consists of multiple webpages that together form the complete system. Each page is developed using HTML for structure, CSS for styling and layout, and JavaScript for interactive behaviour. These tools work together to create a responsive and user-friendly interface.

Homepage (index.html)

Developed by Pravin Raj R P, the home page acts as the entry point of the entire website. The HTML used consists of a navigation bar, content area, and footer area. The prominent part of this page is the cake navigation section, where all the layers of the cake are used as a link to different parts of the site, such as the custom cake page, pre-made products page, and checkout page.

The CSS used in this page is to create a layered effect on the cake, add colour to make it visually appealing, and add effects such as hover effects and shadow effects to make it look three-dimensional.

JavaScript is included primarily for maintaining overall website functionality such as cart updates and shared interactive features across pages.

Pre-Made Cakes and Pastries Page (products.html)

Developed by Aasim Mohammed M S, the products page contains the ready-to-order products from the bakery, such as cakes and pastries. The page is designed using HTML for the product cards, which include the

image, name, and a short description of the product, along with the 'order now' button for adding or removing the product from the cart.

CSS is used for organizing the product cards in a grid view for different screen sizes, ensuring all the products are well ordered. It is also used for adding effects such as hover effects and badges for special products like best sellers and chef's choice.

JavaScript is used for adding interactivity to the products page. It contains filters for different categories like cakes and pastries, which are shown separately to users. JavaScript is also used for the 'order now' button logic for adding products to the cart.

Custom Cake Design Page (custom.html)

Developed by Narendra R, the custom cake page is where users can design custom cakes based on their preferences. The HTML section contains form elements like dropdown menus, radio buttons, and checkboxes, where users can choose their preferred type of cake, flavor, toppings, size, and custom message for their cake.

CSS is used for the design of the custom cake section and for creating the section for the live cake preview, where images of the cake base and toppings are layered using the position attribute of images. The design is such that the custom cake section and the live cake preview section are separately visible for easier usage.

JavaScript is majorly involved in this page, as it is used to dynamically update the cake image based on user selections, as well as determining the price of the cake based on the selected options. It is also used for controlling the cake size selector and enabling or disabling certain inputs based on user selections for customizing the cake.

Cart Page (cart.html)

Developed by Aasim Mohammed M S, the cart page displays the products that the user has added to the cart before they are checked out.

The HTML section displays a list of products added to the cart along with images, names, quantity, and price of the products.

CSS is used for the design of the structured layout where the products in the cart are displayed on one side, and the order summary box shows the total number of products and the total amount of the order.

JavaScript is used for the design of the cart where the products are added from the local storage of the browser, and the quantity of products can be increased or decreased using interactive buttons.

Checkout Page (checkout.html)

Developed by Aatish Ayyapath, the checkout page is the final step of the ordering process. The HTML structure of this page is divided into the following parts: the part for the order summary and the part for the customer information.

CSS is used to divide this page into two parts: the part for the order summary and the part for the payment information. This proper layout helps users review their order and enter required details without confusion.

JavaScript is used to display the items added by the customer to their cart, to compute the price of the items, and to simulate the action of ordering by showing a message upon clicking the ‘Place Order’ button.

Code Samples

The following code snippets show some of the important interactive features implemented in the bakery ordering interface.

1. Product Category Filtering

This code allows users to filter products based on category such as cakes or pastries. When a filter button is clicked, only the relevant products are displayed.

```
JS script.js > ...
/* ========================
   FILTER LOGIC
======================== */
const filterButtons = document.querySelectorAll('.filter-btn');
const products = document.querySelectorAll('.product-card');

if (filterButtons.length && products.length) {
  filterButtons.forEach(btn => {
    btn.addEventListener('click', () => {
      filterButtons.forEach(b => b.classList.remove('active'));
      btn.classList.add('active');

      const category = btn.dataset.category;

      products.forEach(product => {
        product.style.display =
          category === 'all' || product.classList.contains(category)
            ? 'block'
            : 'none';
      });
    });
  });
}
```

2. Cart System Using Local Storage

The cart system stores selected items using the browser's localStorage, allowing the cart data to remain saved even when the user refreshes the page or navigates between different pages.

JS script.js > ...

```javascript
/* ========================
   CART SYSTEM
======================== */

function getCart() {
  return JSON.parse(localStorage.getItem('cart')) || [];
}

function saveCart(cart) {
  localStorage.setItem('cart', JSON.stringify(cart));
}

function findItem(cart, id) {
  return cart.find(item => item.id === id);
}
```

3. Dynamic Cake Preview

The custom cake designer dynamically updates the cake image layers when user selects different toppings.

```
JS script.js > ...
/* MULTI-TOPPING SUPPORT */
toppingInputs.forEach(input => {
  input.addEventListener('change', () => {

    const topping = input.value;

    if (input.checked) {
      toppingLayers[topping].src =
        `images/custom/toppings/${topping}.png`;
      toppingLayers[topping].style.display = "block";
    } else {
      toppingLayers[topping].style.display = "none";
      toppingLayers[topping].src = "";
    }

    calculateCustomPrice();
  });
});
```

4. Custom Cake Price Calculation

The price of the customized cake is calculated dynamically based on flavour, size, toppings, and cake type.

```
JS script.js > ...
function calculateCustomPrice() {
  const flavour = flavourSelect.value;

  if (!basePrices[flavour]) {
    currentCustomPrice = 0;
    document.getElementById("custom-price").innerText = 0;
    return;
  }

  let total = basePrices[flavour] * cakeKg;

  // add egg / eggless price
  const cakeType = document.querySelector('input[name="cake-type"]:checked')?.value;
  if (cakeType) {
    total += cakeTypePrices[cakeType];
  }

  // add topping prices
  toppingInputs.forEach(input => {
    if (input.checked) {
      total += toppingPrices[input.value] || 0;
    }
  });

  currentCustomPrice = total;

  const priceDisplay = document.getElementById("custom-price");
  if (priceDisplay) priceDisplay.innerText = total;
}
```

5. Custom Cake Preview Structure

This HTML structure creates the live cake preview system used in the custom cake designer. Multiple image layers are stacked inside a container so that JavaScript can dynamically update the cake base and toppings.

```html
<!-- PREVIEW -->
<div class="cake-preview">
  <h3>Live Preview</h3>

  <div class="cake-canvas">

    <!-- PLACEHOLDER -->
    <div id="cake-placeholder">Select Your Cake Base</div>

    <!-- BASE -->
    <img id="cake-base" src="" style="display:none;">

    <!-- INDIVIDUAL TOPPING LAYERS -->
    <img id="layer-chips" class="topping-layer">
    <img id="layer-nuts" class="topping-layer">
    <img id="layer-fruits" class="topping-layer">
    <img id="layer-sprinkles" class="topping-layer">

    <!-- TEXT -->
    <div id="cake-text"></div>

  </div>

</div>
```

6. Tooltip Interface

The interface includes tooltip elements that provide additional information to users when they hover over certain options such as cake size, toppings, and cake type.

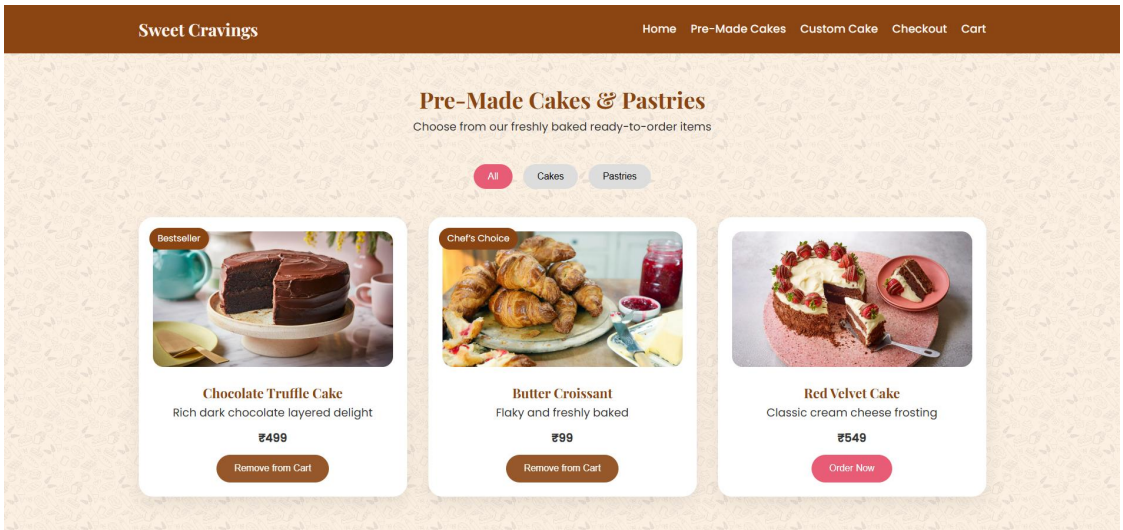
```
css > # style.css > ...

/* ========================
   TOOLTIPS
========================= */

.tooltip {
  position: relative;
  display: inline-block;
  cursor: help;
  margin-left: 5px;
  color: #8B4513;
  font-weight: bold;
}

.tooltip-text {
  visibility: hidden;
  width: 200px;
  background-color: #333;
  color: white;
  text-align: center;
  padding: 8px;
  border-radius: 6px;

  position: absolute;
  bottom: 125%;
  left: 50%;
  transform: translateX(-50%);

  opacity: 0;
  transition: opacity 0.3s;
  font-size: 0.8rem;
}

.tooltip:hover .tooltip-text {
  visibility: visible;
  opacity: 1;
}
```

Screenshots of Output

Figure 1 : Home Page

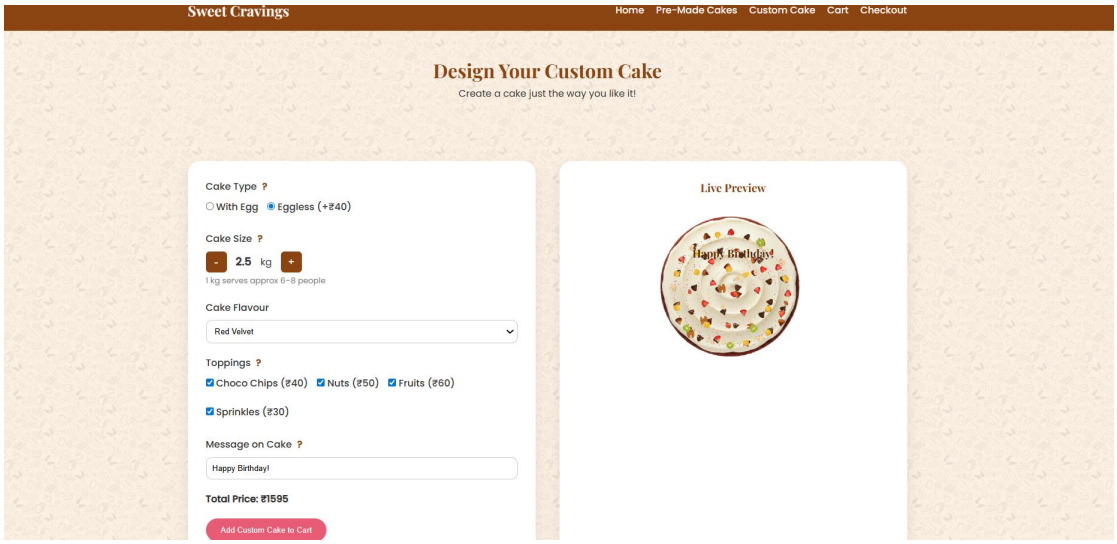
This screenshot shows the homepage of the website. The page includes an interactive cake-shaped navigation interface where each cake layer directs users to different sections of the website as shown in the figure along with a 'Why Choose Us' section.

Figure 2 : Pre-Made Cakes and Pastries Page



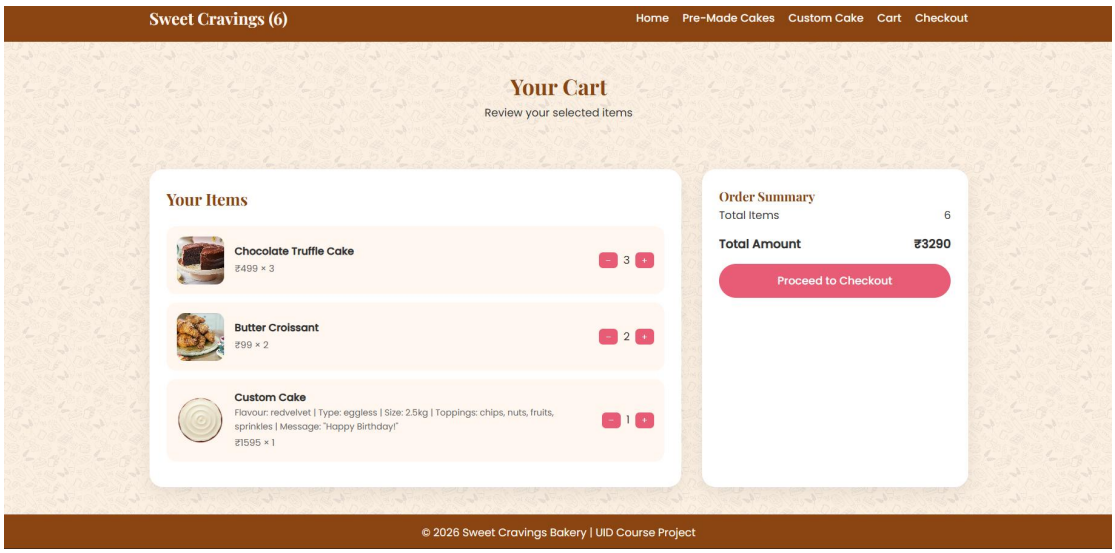
This screenshot displays the pre-made product browsing interface. Each product is shown in a card layout with an image, description, price, and an option to add or remove the item from the cart.

Figure 3 : Custom Cake Design Page



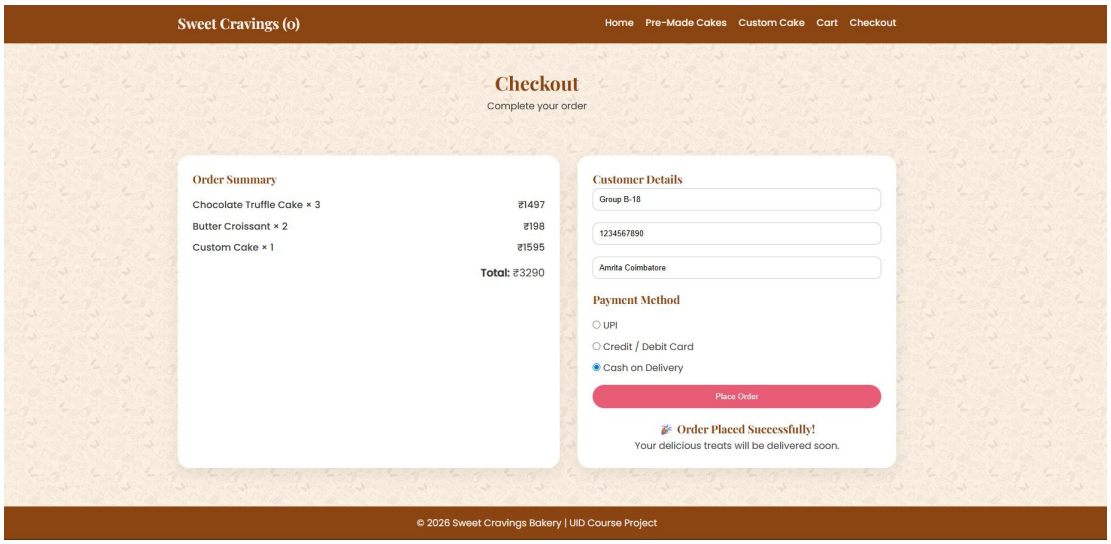
This screenshot shows the custom cake builder page. Users can select cake type, flavour, toppings, size, and custom message. The live preview section dynamically updates to show the selected customization options.

Figure 4 : Cart Section



This screenshot shows the cart interface where users can review the items they have selected. The cart displays product images, item quantities, price details, and the total order amount before proceeding to checkout.

Figure 5: Checkout Page



This screenshot displays the checkout interface where users can review their order summary, enter delivery information, choose a payment method, and place their order.